GATE DA 2027

Calculus and Optimization

Single-variable functions, limits, continuity, differentiability, Taylor series and optimization with the stories of Ramu and Somu

Ramu and Somu once more.
Ramu asks "but why?" about everything. **Somu** answers with everyday pictures: a car's speedometer, a rubber band, a hill walk, a farmer's fence.

Syllabus covered (GATE DA, Section 3, single-variable part): functions of a single variable; limits; continuity and differentiability; Taylor series; optimization of single-variable functions.

Every limit, series, derivative and optimum in these notes was checked with a computer algebra system before it was written down.

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1 Functions of a Single Variable

Ramu & Somu: A vending machine

Somu points at the snack machine. "You press one button, exactly one snack drops. That is a **function**: every input has *exactly one* output." Ramu asks: "Can two buttons give the same snack?" "Yes. Two inputs may share an output, but one input can never give two outputs."

Vocabulary

- A function $f : A \to B$ assigns to each x in the **domain** A exactly one $f(x) \in B$. The **range** is the set of values actually attained.
- **Even**: $f(-x) = f(x)$ (symmetric about the y-axis), e.g. $x^2, \cos x$. **Odd**: $f(-x) = -f(x)$ (symmetric about the origin), e.g. $x^3, \sin x$.
- **Composition**: $(f \circ g)(x) = f(g(x))$; in general $f \circ g \neq g \circ f$.
- **Injective** (one-to-one): $f(a) = f(b) \Rightarrow a = b$. A function has an **inverse** exactly when it is a bijection onto its range; the graph of f^{-1} is the mirror image of the graph of f in the line $y = x$.
- **Monotone**: increasing if $a < b \Rightarrow f(a) \leq f(b)$; strictly increasing if $<$.
- **Periodic**: $f(x + T) = f(x)$.

1.1 The standard functions you must know

Function	Domain → Range	Facts
x^n	$\mathbb{R} \to$ depends on parity of n	even n: even function, ≥ 0; odd n: odd, strictly increasing
$\sqrt{x}$	$[0, \infty) \to [0, \infty)$	inverse of x^2 on $x \geq 0$
$1/x$	$\mathbb{R} \setminus \{0\}$	odd; vertical asymptote $x = 0$
e^x	$\mathbb{R} \to (0, \infty)$	$e^{a+b} = e^a e^b$; strictly increasing; $(e^x)' = e^x$
$\ln x$	$(0, \infty) \to \mathbb{R}$	inverse of e^x; $\ln(ab) = \ln a + \ln b$; $\ln x^k = k \ln x$
$\sin x,\ \cos x$	$\mathbb{R} \to [-1, 1]$	period 2π; $\sin^2 + \cos^2 = 1$
$\tan x$	$x \neq \frac{\pi}{2} + k\pi$	period π; odd
$\lvert x \rvert$	$\mathbb{R} \to [0, \infty)$	even; corner at 0
$\lfloor x \rfloor$	$\mathbb{R} \to \mathbb{Z}$	step function; jumps at integers
$\sigma(x) = \frac{1}{1+e^{-x}}$	$\mathbb{R} \to (0, 1)$	the logistic (sigmoid) function of machine learning; $\sigma' = \sigma(1 - \sigma)$

Domain and even/odd checks

- $f(x) = \sqrt{4 - x^2}$: need $4 - x^2 \geq 0$, so domain $[-2, 2]$, range $[0, 2]$ (upper half circle). It is even.
- $f(x) = \ln(x - 1)$: need $x > 1$, domain $(1, \infty)$.
- $f(x) = x^3 + x$: $f(-x) = -x^3 - x = -f(x)$, odd.
- $f(x) = x^2 + x$: $f(-x) = x^2 - x$, which is neither $f(x)$ nor $-f(x)$, so neither even nor odd.
- Every function is a sum of an even and an odd part: $f = \frac{f(x)+f(-x)}{2} + \frac{f(x)-f(-x)}{2}$.

Common GATE traps

(1) Domain first! $\frac{x^2-1}{x-1}$ equals $x+1$ only for $x \neq 1$. (2) $\sqrt{x^2} = |x|$, not x. (3) $\ln(a+b) \neq \ln a + \ln b$. (4) $(f \circ g) \neq (g \circ f)$ in general. (5) A function that is not injective has no inverse until you restrict its domain (for example x^2 on $x \geq 0$).

2 Limits

Ramu & Somu: Walking toward a wall

Ramu walks toward a wall and halves the distance each step. "Do I ever touch it?" Somu: "Never exactly, but you get as close as you like. The wall is the **limit** of your position. A limit describes where a function is *heading* as x approaches a point, never what happens *at* the point." A hole in the road at the wall does not change where you were heading.

Definition and one-sided limits

$\lim_{x \to a} f(x) = L$ means: for every $\varepsilon > 0$ there is $\delta > 0$ with $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$ (stay within δ of a, and f stays within ε of L). The value $f(a)$ itself plays no role.

The limit exists *iff* both one-sided limits exist and are equal: $\lim_{x \to a^-} f = \lim_{x \to a^+} f = L$.

Limit laws (when the individual limits exist)

- $\lim(f \pm g) = \lim f \pm \lim g$; $\lim(fg) = \lim f \cdot \lim g$; $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ if $\lim g \neq 0$.
- $\lim cf = c \lim f$; $\lim f^n = (\lim f)^n$.
- **Squeeze theorem:** if $g \leq f \leq h$ near a and $\lim g = \lim h = L$, then $\lim f = L$.
- Continuous outer function: $\lim g(f(x)) = g(\lim f(x))$.

2.1 The seven indeterminate forms

Idea in plain words

$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0 are *not numbers*; they only say "the naive substitution failed, you must do more work". For example $\frac{0}{0}$ can be 1 (as in $\frac{\sin x}{x}$), 0 (as in $\frac{x^2}{x}$) or ∞ (as in $\frac{x}{x^2}$).

2.2 Techniques, in the order you should try them

1. **Substitute.** If the function is continuous at a the answer is $f(a)$.
2. **Factor and cancel** ($\frac{0}{0}$ with polynomials): $\lim_{x \to 1} \dfrac{x^2 - 1}{x - 1} = \lim(x + 1) = 2$.
3. **Rationalise** (roots): $\lim_{x \to \infty} \left(\sqrt{x^2 + x} - x\right) = \lim \dfrac{x}{\sqrt{x^2 + x} + x} = \dfrac{1}{2}$.
4. **Divide by the highest power** (limits at infinity): $\lim_{x \to \infty} \dfrac{3x^2 + 2x}{5x^2 + 1} = \dfrac{3}{5}$.
5. **Standard limits** (below).
6. **L'Hôpital's rule** or **Taylor series**.

Standard limits to memorise

$$\lim_{x\to 0}\frac{\sin x}{x}=1 \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2} \qquad \lim_{x\to 0}\frac{\tan x}{x}=1$$

$$\lim_{x\to 0}\frac{e^x-1}{x}=1 \qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1 \qquad \lim_{x\to 0}\frac{a^x-1}{x}=\ln a$$

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e \qquad \lim_{x\to 0}(1+x)^{1/x}=e \qquad \lim_{x\to 0}\frac{(1+x)^n-1}{x}=n$$

$$\lim_{x\to\infty}\frac{\ln x}{x^p}=0\ (p>0) \qquad \lim_{x\to\infty}\frac{x^p}{e^x}=0 \qquad \lim_{x\to 0^+}x^p\ln x=0\ (p>0)$$

$$\lim_{x\to 0^+}x^x=1 \qquad \lim_{x\to\infty}\frac{\sin x}{x}=0 \text{ (squeeze between } \pm\tfrac{1}{x})$$

The growth order is: $\ln x \ll x^p \ll a^x \ll x!$ $(a>1)$. Exponentials beat polynomials, and polynomials beat logarithms.

Ramu & Somu: Why $\frac{\sin x}{x}\to 1$

Ramu is puzzled: both $\sin x$ and x go to 0, so why does the ratio settle at 1? Somu draws a unit circle: for a tiny angle x (in radians) the arc length is x and the height is $\sin x$, and for small angles the arc and the vertical height are almost the same length. So the ratio $\to 1$. This only works in **radians**; in degrees the limit is $\frac{\pi}{180}$.

2.3 L'Hôpital's rule

Rule

If $\lim \frac{f}{g}$ has the form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ and $\lim \frac{f'}{g'}$ exists (or is $\pm\infty$), then

$$\lim\frac{f(x)}{g(x)}=\lim\frac{f'(x)}{g'(x)}.$$

Other forms are first converted: $0\cdot\infty \to \frac{f}{1/g}$; $\infty-\infty \to$ common denominator or rationalise; $1^\infty, 0^0, \infty^0 \to$ take ln first: if $y=f^g$ then $\ln y = g\ln f$ and $\lim y = e^{\lim \ln y}$.

Six limits worked out

1. $\lim_{x\to 0}\frac{e^x-1-x}{x^2}$: $\frac{0}{0}$. Differentiate: $\frac{e^x-1}{2x}$, still $\frac{0}{0}$; again: $\frac{e^x}{2}\to\frac{1}{2}$. **Answer $\frac{1}{2}$.**

2. $\lim_{x\to 0}\frac{\sin 3x}{\sin 5x}$: use $\frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x}\cdot\frac{3}{5}\to\frac{3}{5}$. **Answer $\frac{3}{5}$.**

3. $\lim_{x\to 0^+} x\ln x$: $0\cdot(-\infty)$. Write $\frac{\ln x}{1/x}$ $(\frac{-\infty}{\infty})$; L'Hôpital gives $\frac{1/x}{-1/x^2}=-x\to 0$. **Answer 0.**

4. $\lim_{x\to 0}(1+2x)^{1/x}$: 1^∞. $\ln y=\frac{\ln(1+2x)}{x}\to 2$. So $y\to e^2$. **Answer e^2.**

5. $\lim_{x\to 0^+} x^x$: 0^0. $\ln y = x\ln x\to 0$, so $y\to e^0=1$. **Answer 1.**

6. $\lim_{x\to 0}(\cos x)^{1/x^2}$: $\ln y=\frac{\ln\cos x}{x^2}\to\frac{-\tan x}{2x}\to-\frac{1}{2}$. **Answer $e^{-1/2}$.**

2.4 Limits at infinity and asymptotes

- **Horizontal asymptote** $y = L$ if $\lim_{x\to\pm\infty} f = L$. For a rational function, compare degrees: $\deg p < \deg q \Rightarrow 0$; equal $\Rightarrow$ ratio of leading coefficients; $\deg p > \deg q \Rightarrow$ no horizontal asymptote.
- **Vertical asymptote** at $x = a$ if some one-sided limit is $\pm\infty$.
- **Slant asymptote** $y = mx + c$ with $m = \lim \frac{f(x)}{x}$, $c = \lim(f(x) - mx)$. Example: $f = \frac{2x^2+1}{x-1}$ has $m = 2$ and $c = \lim(\frac{2x^2+1}{x-1} - 2x) = 2$, so $y = 2x + 2$.

One-sided limits disagree

$\lim_{x\to 0} \frac{|x|}{x}$: from the right it is $+1$, from the left -1, so the limit **does not exist**. Likewise $e^{1/x} \to \infty$ as $x \to 0^+$ but $\to 0$ as $x \to 0^-$.

Common GATE traps

(1) L'Hôpital needs a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form *first*; otherwise it gives wrong answers. (2) You differentiate numerator and denominator *separately*; it is not the quotient rule. (3) $\lim \frac{\sin x}{x} = 1$ at 0, but $\to 0$ at ∞. (4) $\lim_{x\to\infty}(1 + \frac{1}{x})^x = e$, not 1 and not ∞. (5) If the limit of $\frac{f'}{g'}$ does not exist, the rule says nothing: try something else. (6) A limit may exist even when $f(a)$ is undefined.

3 Continuity

Ramu & Somu: Drawing without lifting the pen

"A function is continuous," says Somu, "if you can draw its graph without lifting your pencil." Ramu objects that a mathematician needs something sharper than a pencil. So Somu states it with limits: at the point a three things must hold, namely $f(a)$ exists, $\lim_{x\to a} f(x)$ exists, and they are *equal*.

Continuity at a point

f is continuous at a iff $\lim_{x\to a} f(x) = f(a)$. Equivalently $\lim_{x\to a^-} f = \lim_{x\to a^+} f = f(a)$. A function is continuous on an interval if it is continuous at every point of it. Polynomials, $e^x, \sin x, \cos x$ are continuous everywhere; rational functions, $\tan x$, $\ln x$ and $\sqrt{x}$ are continuous on their domains; sums, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous.

3.1 Types of discontinuity

Type	What happens	Example
Removable	limit exists, but $\neq f(a)$ or $f(a)$ undefined; can be repaired by redefining $f(a)$	$\frac{x^2-4}{x-2}$ at 2 (limit 4)
Jump	left and right limits exist but differ	$\lfloor x \rfloor$ at integers; $\frac{\lvert x \rvert}{x}$ at 0
Infinite	a one-sided limit is $\pm\infty$	$\frac{1}{x}$ at 0
Oscillating	no limit because of endless wiggling	$\sin\frac{1}{x}$ at 0

Finding a parameter

Let $f(x) = x^2$ for $x < 1$ and $f(x) = ax + 3$ for $x \geq 1$. Continuity at $x = 1$: left limit $= 1$, value from the right $= a + 3$. Setting $1 = a + 3$ gives $a = -2$.

3.2 Two big theorems for continuous functions on a closed interval

IVT and EVT

- **Intermediate Value Theorem.** If f is continuous on $[a, b]$ and k lies between $f(a)$ and $f(b)$, then $f(c) = k$ for some $c \in (a, b)$. Use: showing a root exists (if $f(a)f(b) < 0$, there is a root in between).
- **Extreme Value Theorem.** A continuous function on a *closed bounded* interval $[a, b]$ attains an absolute maximum and an absolute minimum.

A root by IVT

$f(x) = x^3 - x - 1$ is continuous, $f(1) = -1 < 0$ and $f(2) = 5 > 0$. So there is a root in $(1, 2)$. Evaluate $f(1.5) = 0.875 > 0$: the root lies in $(1, 1.5)$. Repeating this halving is the **bisection method**.

Common GATE traps

(1) The EVT can fail on open or unbounded intervals: $f(x) = x$ on $(0, 1)$ has neither a max nor a min; $\frac{1}{x}$ on $(0, 1]$ has no maximum. (2) The IVT needs continuity: $f(x) = \frac{1}{x}$ goes from -1 to 1 on $[-1, 1]$ without ever being 0. (3) Continuity at a point does *not* require continuity nearby, but all standard exam functions are well behaved.

4 Differentiability

Ramu & Somu: The speedometer

Ramu drives from Hyderabad to Vijayawada in 4 hours, covering 280 km. "My average speed is 70 km/h. But the speedometer on the dashboard shows something different every moment!" Somu: "That dashboard reading is the **derivative**: the speed at one instant. It is the average speed over a time interval that shrinks to zero."

Derivative

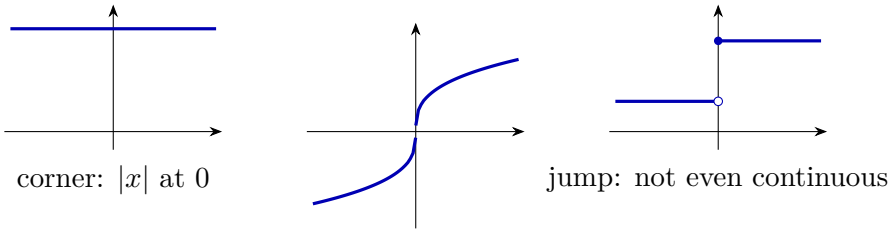
$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}.$$

Geometrically: the slope of the tangent line at $x = a$. The tangent line is $y = f(a) + f'(a)(x-a)$. f is **differentiable** at a when this limit exists (a finite number). One-sided versions: left derivative ($h \to 0^-$) and right derivative ($h \to 0^+$); the derivative exists iff both exist and are equal.

The key relation

Differentiable ⇒ continuous. (Proof: $f(x) - f(a) = \frac{f(x)-f(a)}{x-a} \cdot (x-a) \to f'(a) \cdot 0 = 0$.)
The converse is false: continuity does not give differentiability.

4.1 How differentiability fails



corner: $|x|$ at 0 jump: not even continuous

vertical tangent: $x^{1/3}$ at 0

Standard cases

- $f(x) = |x|$ at 0: right derivative $+1$, left derivative -1; not differentiable (corner), but continuous.
- $f(x) = x^{1/3}$ at 0: the difference quotient is $h^{-2/3} \to \infty$; vertical tangent, not differentiable.
- $f(x) = x|x|$ at 0: the quotient is $|h| \to 0$, so $f'(0) = 0$. Differentiable once (not twice).
- $f(x) = x^2 \sin \frac{1}{x}$ ($f(0) = 0$): $\frac{f(h)}{h} = h \sin \frac{1}{h} \to 0$, so $f'(0) = 0$, yet $f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$ has no limit at 0. So f' exists everywhere but is *not continuous*.
- $f(x) = x \sin \frac{1}{x}$ ($f(0) = 0$): continuous at 0 but $\frac{f(h)}{h} = \sin \frac{1}{h}$ oscillates, so not differentiable.

4.2 Derivative rules

Rules

Sum, constant	$(af + bg)' = af' + bg'$
Product	$(fg)' = f'g + fg'$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
Chain	$\big(f(g(x))\big)' = f'(g(x)) \cdot g'(x)$
Inverse	$(f^{-1})'(y) = \frac{1}{f'(x)}$ where $y = f(x)$
Implicit	differentiate both sides w.r.t. x, treat y as $y(x)$
Logarithmic	for $y = f^g$: $\ln y = g \ln f$, then differentiate

Table of derivatives

$(x^n)' = nx^{n-1}$	$(e^x)' = e^x$	$(a^x)' = a^x \ln a$	$(\ln x)' = \frac{1}{x}$
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(\tan x)' = \sec^2 x$	$(\sec x)' = \sec x \tan x$
$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$	$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$	$(\arctan x)' = \frac{1}{1+x^2}$	$(\sinh x)' = \cosh x$

Derivatives that need care

1. $y = x^x$ $(x > 0)$: $\ln y = x \ln x$, so $\frac{y'}{y} = \ln x + 1$ and $y' = x^x(\ln x + 1)$.
2. $y = xe^{-x}$: $y' = (1 - x)e^{-x}$.
3. $y = \frac{\ln x}{x}$: $y' = \frac{1 - \ln x}{x^2}$.
4. $y = \sigma(x) = \frac{1}{1+e^{-x}}$: $y' = \sigma(1 - \sigma)$. Maximum slope $\frac{1}{4}$ at $x = 0$.
5. $y = \arctan x$: $y' = \frac{1}{1+x^2}$.
6. $y = x \ln x$: $y' = \ln x + 1$.

4.3 Rolle's theorem and the Mean Value Theorem

Ramu & Somu: The speed camera

A speed camera at the start and another at the end of Ramu's trip recorded only his *average* speed of 70 km/h. Somu: "At some moment on the way your speedometer read *exactly* 70." That is the **Mean Value Theorem**: the instantaneous rate must equal the average rate somewhere.

Theorems

- **Rolle:** if f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.
- **MVT:** if f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.
- **Consequences:** $f' > 0$ on an interval $\Rightarrow$ strictly increasing; $f' = 0$ on an interval $\Rightarrow$ constant; $|f'| \le M \Rightarrow |f(x) - f(y)| \le M|x - y|$ (Lipschitz).

Finding c

$f(x) = x^2$ on $[1, 3]$: average slope $\frac{9-1}{2} = 4$, and $f'(c) = 2c = 4$ gives $c = 2$, which lies in $(1, 3)$. For $f(x) = x^3 - 3x$ on $[-\sqrt{3}, \sqrt{3}]$: $f(\pm\sqrt{3}) = 0$, and $f'(x) = 3x^2 - 3 = 0$ gives $c = \pm 1$ (Rolle).

Common GATE traps

(1) Differentiable $\Rightarrow$ continuous but not the converse; $|x|$ is the standard witness. (2) MVT fails for $|x|$ on $[-1, 1]$: average slope 0 but f' is never 0 (no derivative at 0), because the hypothesis fails. (3) $f'(a) = 0$ does *not* imply a maximum or minimum (x^3 at 0). (4) The chain rule applies from the outside in: $(\sin x^2)' = 2x \cos x^2$, not $\cos x^2$.

5 Taylor Series

Ramu & Somu: Zooming into a curvy road

Ramu stands on a hilly road and asks Somu to describe it using only what he sees *right here*. Somu: "First I say: the road is at height $f(a)$. Better: it also slopes by $f'(a)$, so a straight line. Better still: it bends by $f''(a)$, so a parabola. Each extra derivative I know adds one more term and lets me predict the road a little farther away." That is a **Taylor polynomial**: it copies the function's height, slope, bend, and so on at one point.

Taylor and Maclaurin series

If f has derivatives of all orders at a,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

The special case $a = 0$ is the **Maclaurin series**. The **Taylor polynomial of degree** n is the sum up to $k = n$; it is the *unique* polynomial of degree $\le n$ that matches f and its first n derivatives at a.

Lagrange remainder: how big is the error?

$f(x) = P_n(x) + R_n(x)$ with

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } x.$$

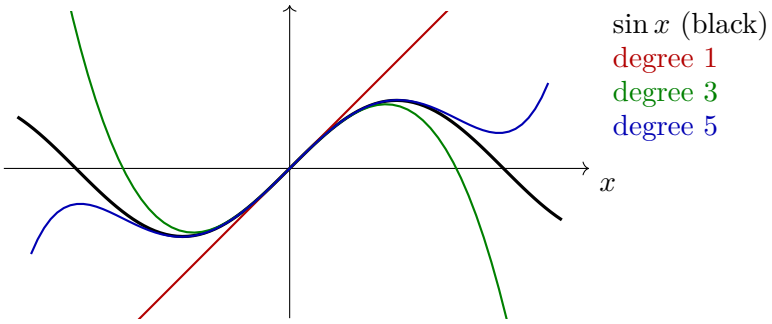
So $|R_n| \le \dfrac{M}{(n+1)!}|x-a|^{n+1}$, where M bounds $|f^{(n+1)}|$ between a and x. The error shrinks quickly when x is close to a or n is large. For $n = 0$ this is exactly the Mean Value Theorem.

5.1 Standard Maclaurin series (memorise these)

Function	Series	Valid for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum \frac{x^k}{k!}$	all x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	all x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	all x
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \cdots$	$\lvert x\rvert < 1$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$-1 < x \le 1$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	$\lvert x\rvert \le 1$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \cdots$	$\lvert x\rvert < 1$
$\sqrt{1+x}$	$1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots$	$\lvert x\rvert \le 1$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$	$\lvert x\rvert < \frac{\pi}{2}$

Idea in plain words

Patterns that save time. *Even* functions ($\cos x$) contain only even powers and *odd* functions ($\sin x, \arctan x$) only odd powers. $\sin x$ and $\cos x$ alternate signs because their derivatives cycle $\sin \to \cos \to -\sin \to -\cos$. $\frac{1}{1-x}$ is the geometric series; replacing x by $-x$ and integrating gives $\ln(1+x)$.



Near $x = 0$ all the polynomials hug $\sin x$; farther away the low-degree ones fly off while the degree-5 curve stays accurate longer.

5.2 Worked examples

1. Taylor polynomial about a point other than 0

$f(x) = x^3 - 2x + 1$ about $a = 1$: $f(1) = 0$, $f'(x) = 3x^2 - 2 \Rightarrow f'(1) = 1$, $f''(x) = 6x \Rightarrow f''(1) = 6$, $f''' = 6$. So

$$f(x) = (x-1) + 3(x-1)^2 + (x-1)^3,$$

and the series stops: a polynomial is its own Taylor series.

2. Approximations with error bounds

- $e^{0.5}$ with degree 3: $1 + 0.5 + 0.125 + 0.0208333 = 1.6458333$ (true value 1.6487213). Error bound: $\frac{e^{0.5}}{4!}(0.5)^4 \le 0.0043$; the actual error 0.0029 is within the bound.
- $\sqrt{1.1}$ with degree 2 ($x = 0.1$): $1 + 0.05 - 0.00125 = 1.04875$ (true 1.0488088).
- $\sin 0.1$ with degree 3: $0.1 - \frac{0.001}{6} = 0.0998333$ (true 0.0998334).
- $\ln 1.2$ with degree 3: $0.2 - 0.02 + 0.00267 = 0.18267$ (true 0.18232).

3. Limits by Taylor series: often faster than L'Hôpital

- $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$: $x - \sin x = \frac{x^3}{6} - \cdots$, so the limit is $\frac{1}{6}$.
- $\lim_{x\to 0} \dfrac{\cos x - 1 + \frac{x^2}{2}}{x^4} = \dfrac{1}{24}$ (the next term of $\cos x$).
- $\lim_{x\to 0} \dfrac{\tan x - x}{x^3} = \dfrac{1}{3}$ (from $\tan x = x + \frac{x^3}{3} + \cdots$).
- $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2} = \dfrac{1}{2}$.

4. Building new series from old ones

- **Substitute:** $e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \cdots$ (replace x by $-x^2$ in e^x).
- **Multiply:** $e^x \sin x = (1 + x + \frac{x^2}{2} + \cdots)(x - \frac{x^3}{6} + \cdots) = x + x^2 + \frac{x^3}{3} + \cdots$
- **Differentiate term by term:** $(\sin x)' = \cos x$ reproduces the cosine series.
- **Integrate term by term:** integrating $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ gives $\arctan x$.
- The coefficient of x^k in a Maclaurin series equals $\frac{f^{(k)}(0)}{k!}$. So $f^{(5)}(0)$ for $\sin x$ is $5! \cdot \frac{1}{5!} = 1$, and $f^{(4)}(0)$ for e^x is 1.

5.3 Radius of convergence

Ratio test

For $\sum a_k (x-a)^k$ the **radius of convergence** is $R = \lim_{k\to\infty} \left|\frac{a_k}{a_{k+1}}\right|$. The series converges for $|x-a| < R$, diverges for $|x-a| > R$, and the endpoints must be checked separately. Examples: $\sum \frac{x^k}{k!}$: $R = \infty$; $\sum x^k$: $R = 1$; $\sum \frac{x^k}{k}$ $(= -\ln(1-x))$: $R = 1$, converges at $x = -1$, diverges at $x = 1$.

Common GATE traps

(1) Don't forget the factorial: the coefficient is $\frac{f^{(k)}(a)}{k!}$. (2) The series about a uses powers of $(x-a)$, not x. (3) A function can have a Taylor series that converges but not to the function (e.g. e^{-1/x^2} has all derivatives 0 at 0). (4) $\ln(1+x)$'s series needs $|x| \le 1$, so you cannot use it directly for $\ln 3$. (5) The nth-degree remainder involves the $(n+1)$st derivative.

6 Optimization of Single-variable Functions

Ramu & Somu: Walking a hill in the fog

Ramu and Somu climb a foggy hill. "How do we know we are at a peak?" Somu: "At the top the ground is level for a moment: slope $= 0$. Before the top you were walking uphill (slope > 0), after it downhill (slope < 0)." Ramu: "But a flat spot could also be a bench halfway up!" "Right, so we also look at how the slope itself changes: at a peak it keeps decreasing, so the second derivative is negative."

6.1 Terminology

Definitions

- x^* is a **local (relative) maximum** if $f(x^*) \ge f(x)$ for all x near x^*; **global (absolute) maximum** if it holds for all x in the domain. Minima are defined likewise.
- A **critical point** is an interior point where $f'(x) = 0$ or $f'(x)$ does not exist.
- **Fermat's theorem:** if f has a local extremum at an interior point where it is differentiable, then $f'(x^*) = 0$. (So extrema hide among critical points and end points.)
- f is **concave up** (convex) where $f'' > 0$ and **concave down** where $f'' < 0$. An **inflection point** is where the concavity changes.

6.2 The tests

First-derivative test

At a critical point c: if f' changes from $+$ to $-$, c is a local maximum; from $-$ to $+$, a local minimum; no sign change, neither.

Second-derivative test

At a critical point with $f'(c) = 0$: $f''(c) > 0 \Rightarrow$ local minimum; $f''(c) < 0 \Rightarrow$ local maximum; $f''(c) = 0 \Rightarrow$ **inconclusive** (use the first-derivative test or higher derivatives).
Why: near c, Taylor gives $f(c+h) \approx f(c) + \frac{f''(c)}{2}h^2$. If $f''(c) > 0$ every nearby value is larger than $f(c)$: a valley.

Higher-order test

If $f'(c) = \cdots = f^{(n-1)}(c) = 0$ and $f^{(n)}(c) \neq 0$: for **even** n it is a local min ($f^{(n)} > 0$) or max ($f^{(n)} < 0$); for **odd** n it is an inflection point (no extremum). Examples: x^4 at 0 ($n = 4$, minimum, although $f''(0) = 0$); x^3 at 0 ($n = 3$, inflection).

Closed-interval method (global extrema of continuous f on $[a, b]$)

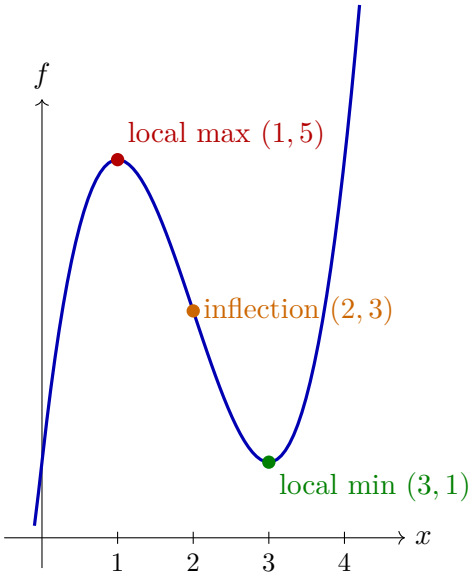
Evaluate f at (i) the end points a, b and (ii) every critical point inside; the largest value is the global maximum and the smallest the global minimum. (The EVT guarantees they exist.)

Full analysis of $f(x) = x^3 - 6x^2 + 9x + 1$

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, so critical points $x = 1, 3$. $f''(x) = 6x - 12$.

interval	$x < 1$	$1 < x < 3$	$x > 3$
sign of f'	$+$	$-$	$+$
behaviour	increasing	decreasing	increasing

$f''(1) = -6 < 0$: local **max** $f(1) = 5$. $f''(3) = 6 > 0$: local **min** $f(3) = 1$. Inflection at $f'' = 0 \Rightarrow x = 2$.
On $[0, 4]$: $f(0) = 1$, $f(1) = 5$, $f(3) = 1$, $f(4) = 5$. So the global max is 5 (at $x = 1$ and $x = 4$) and the global min is 1 (at $x = 0$ and $x = 3$).



Tricky critical points

- $f(x) = xe^{-x}$: $f' = (1-x)e^{-x} = 0 \Rightarrow x = 1$; f' goes $+ \to -$: maximum $f(1) = \frac{1}{e} \approx 0.368$.
- $f(x) = x \ln x$ $(x > 0)$: $f' = \ln x + 1 = 0 \Rightarrow x = \frac{1}{e}$; $f'' = \frac{1}{x} > 0$: minimum $f(\frac{1}{e}) = -\frac{1}{e}$. (Used in the entropy function of information theory.)
- $f(x) = \frac{\ln x}{x}$: $f' = \frac{1-\ln x}{x^2} = 0 \Rightarrow x = e$: maximum $\frac{1}{e}$. (So $e^{1/e}$ is the largest value of $x^{1/x}$.)
- $f(x) = x + \frac{1}{x}$ for $x > 0$: $f' = 1 - \frac{1}{x^2} = 0 \Rightarrow x = 1$, minimum 2 (AM–GM).
- $f(x) = \sin x + \cos x$: maximum $\sqrt{2}$ at $x = \frac{\pi}{4}$.
- $f(x) = x^3 - 3x$ on $[-2, 3]$: values $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Global max 18 at $x = 3$; global min -2 at $x = -2$ and $x = 1$.
- $f(x) = |x|$: minimum at 0 where f' does not exist: a critical point of the second kind.
- $f(x) = x^4 - x^2 + 1$: $f' = 4x^3 - 2x = 2x(2x^2 - 1) = 0 \Rightarrow x = 0, \pm\frac{1}{\sqrt{2}}$. Minima at $\pm\frac{1}{\sqrt{2}}$ with value $\frac{3}{4}$; local maximum 1 at $x = 0$.

6.3 Word problems: the recipe

Idea in plain words

(1) Draw and name the quantities. (2) Write the **objective** (what to maximise or minimise). (3) Write the **constraint** and use it to eliminate all but one variable. (4) State the domain. (5) Find critical points and compare with the end points. (6) Answer the question that was asked (units, the second quantity).

Four classics

1. **Rectangle with perimeter** 40**.** $A = x(20 - x)$, $A' = 20 - 2x = 0 \Rightarrow x = 10$; the maximum area is 100 (a square).

2. **Farmer with 100 m of fence along a river** (no fence needed on the river side): $A = x(100 - 2x)$, $A' = 100 - 4x = 0 \Rightarrow x = 25$; width 25, length 50; area 1250 m^2.

3. **Open box from a** 12×12 **sheet** by cutting squares of side s from the corners: $V = s(12-2s)^2$, $V' = (12-2s)(12-6s) = 0 \Rightarrow s = 2$ (since $s = 6$ gives $V = 0$). Maximum volume $V(2) = 128$.

4. **Closed cylindrical can of volume** 1000 **cm**^{3} with the least material: $A = 2\pi r^2 + \frac{2000}{r}$,

$A' = 4\pi r - \frac{2000}{r^2} = 0 \Rightarrow r^3 = \frac{500}{\pi}$, $r \approx 5.42$ cm, and then $h = \frac{1000}{\pi r^2} = 2r$. The best can has height equal to its diameter.

5. **Closest point on $y = x^2$ to $(0, 1)$.** Minimise $d^2 = x^2 + (x^2 - 1)^2 = x^4 - x^2 + 1$ (squaring avoids the root and has the same minimiser). Minimum at $x^2 = \frac{1}{2}$: $d^2 = \frac{3}{4}$, so $d = \frac{\sqrt{3}}{2}$.

6.4 Convexity: why optimisers love it

Convex functions

f is **convex** on an interval if $f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y)$ for all $0 \le \lambda \le 1$: every chord lies above the graph. For twice-differentiable f this is equivalent to $f'' \ge 0$. For convex f:

- every critical point is a **global minimum** (no local-versus-global worry); $f(y) \ge f(x) + f'(x)(y - x)$: the tangent line lies below the graph;
- Jensen's inequality: $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$.

Examples of convex functions: x^2, e^x, $|x|$, $-\ln x$, $x \ln x$. Squared-error loss and logistic loss in machine learning are convex, which is why training them is reliable.

6.5 Numerical methods that come from calculus

Ramu & Somu: Rolling a ball downhill

Ramu is told to find the lowest point of $f(x) = (x - 3)^2$ without algebra. Somu: "Stand anywhere, look at the slope, and step *against* it: $x_{k+1} = x_k - \eta f'(x_k)$. The number $\eta > 0$ is the **learning rate**." This is **gradient descent**.

For $f = (x - 3)^2$ we have $f' = 2(x - 3)$, so $x_{k+1} - 3 = (1 - 2\eta)(x_k - 3)$. Starting from $x_0 = 0$ with $\eta = 0.1$ (factor 0.8):

k	0	1	2	3	4	5
x_k	0	0.6	1.08	1.464	1.7712	2.017

The iterates creep toward 3. The method converges iff $|1 - 2\eta| < 1$, i.e. $0 < \eta < 1$; for $\eta = 1$ it bounces between 0 and 6 forever, and for $\eta > 1$ it diverges. In general the step size must be below $\frac{2}{L}$ where L bounds f''.

Newton's method

Root finding: $x_{k+1} = x_k - \dfrac{f(x_k)}{f'(x_k)}$ (follow the tangent line to the axis). Example $f = x^2 - 2$, $x_0 = 1$: $x_1 = 1.5$, $x_2 = 1.41667$, $x_3 = 1.41422$, $x_4 = 1.41421 \to \sqrt{2}$. The number of correct digits roughly *doubles* each step (quadratic convergence).

Optimisation: apply it to f': $x_{k+1} = x_k - \dfrac{f'(x_k)}{f''(x_k)}$; it uses the curvature to choose the step. For a quadratic it lands on the minimum in one step.

Common GATE traps

(1) $f'(c)=0$ does not guarantee an extremum (x^3). (2) $f''(c)=0$ does not rule one out (x^4). (3) Global extrema on a closed interval may be at the end points, where $f'\neq 0$. (4) On an open interval a maximum may not exist at all. (5) Check the domain: $x\ln x$ needs $x>0$. (6) In word problems verify that the critical point is really a max or min and not a boundary or a zero-volume case. (7) A local min of a non-convex function need not be a global min.

7 One-page Cheat Sheet

Limits

Standard: $\frac{\sin x}{x}, \frac{\tan x}{x}\to 1$; $\frac{1-\cos x}{x^2}\to\frac{1}{2}$; $\frac{e^x-1}{x}, \frac{\ln(1+x)}{x}\to 1$; $(1+\frac{1}{x})^x\to e$; $x^x\to 1$ (0^+); $x\ln x\to 0$ (0^+). Growth: $\ln x \ll x^p \ll a^x \ll x!$. L'Hôpital only for $\frac{0}{0}, \frac{\infty}{\infty}$. Exponential forms: take ln.

Continuity and differentiability

Continuous at a: $\lim_{x\to a} f = f(a)$. Differentiable $\Rightarrow$ continuous (not conversely). Fails at corners ($|x|$), cusps and vertical tangents ($x^{1/3}$), jumps and oscillation. Rolle: $f(a)=f(b)\Rightarrow f'(c)=0$. MVT: $f'(c)=\frac{f(b)-f(a)}{b-a}$. IVT, EVT need continuity on a closed interval.

Taylor

$f(x)=\sum \frac{f^{(k)}(a)}{k!}(x-a)^k$; remainder $R_n=\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$. e^x, $\sin x$, $\cos x$ converge everywhere; $\frac{1}{1-x}$, $\ln(1+x)$, $(1+x)^\alpha$ for $|x|<1$. Radius $R=\lim|a_k/a_{k+1}|$.

Optimization

Critical points: $f'=0$ or undefined. $f''>0$: min; $f''<0$: max; $f''=0$: use the first-derivative or higher-order test. Closed interval: compare end points and critical points. Convex ($f''\geq 0$): every critical point is a global minimum. Gradient descent $x\leftarrow x-\eta f'(x)$; Newton $x\leftarrow x-\frac{f'}{f''}$.

8 Practice Questions (with answers)

Try each one yourself first. Every answer was checked with a computer algebra system.

1. $\lim\limits_{x\to\infty}\dfrac{3x^2+2x}{5x^2+1}$. **Answer:** $\frac{3}{5}$.

2. $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x\tan x}$. **Answer:** $\frac{1}{2}$.

3. $\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}$. **Answer:** $\frac{1}{6}$.

4. $\lim\limits_{x\to 2}\dfrac{x^3-8}{x-2}$. **Answer:** 12 (this is $f'(2)$ for $f=x^3$).

5. $\lim\limits_{x\to\infty}\left(1+\dfrac{3}{x}\right)^x$. **Answer:** e^3.

6. $\lim_{x\to\infty} x\sin\dfrac{1}{x}$. **Answer:** 1 (put $t=\frac{1}{x}$).
7. $\lim_{x\to 0}\dfrac{\sin 3x}{\sin 5x}$. **Answer:** $\frac{3}{5}$.
8. For which k is $f(x)=\frac{\sin 2x}{x}$ $(x\neq 0)$, $f(0)=k$ continuous? **Answer:** $k=2$.
9. Does $\lim_{x\to 0}\dfrac{|x|}{x}$ exist? **Answer: No** (left -1, right $+1$).
10. Is $f(x)=|x-1|+|x-2|$ differentiable at $x=1$? At $x=1.5$? **Answer: not at** 1 **(corner); yes at** 1.5 (slope 0).
11. $f(x)=x^2\sin\frac{1}{x}$, $f(0)=0$. Value of $f'(0)$? Is f' continuous at 0? **Answer:** $f'(0)=0$**;** f' **is not continuous at** 0.
12. Derivative of x^x at $x=1$. **Answer:** 1 $(x^x(\ln x+1))$.
13. MVT point for $f=x^2$ on $[1,3]$. **Answer:** $c=2$.
14. Does $x^3-x-1=0$ have a root in $(1,2)$? **Answer: Yes** (IVT: $f(1)=-1$, $f(2)=5$).
15. Coefficient of x^3 in the Maclaurin series of $e^x\sin x$. **Answer:** $\frac{1}{3}$.
16. Degree-2 Maclaurin approximation of $\sqrt{1+x}$ and its value at 0.1. **Answer:** $1+\frac{x}{2}-\frac{x^2}{8}$**;** 1.04875.
17. Taylor series of x^3-2x+1 about $x=1$. **Answer:** $(x-1)+3(x-1)^2+(x-1)^3$.
18. Radius of convergence of $\sum\frac{x^k}{k}$. **Answer:** 1.
19. Maximum of $f(x)=xe^{-x}$. **Answer:** $\frac{1}{e}$ **at** $x=1$.
20. Global max and min of $f(x)=x^3-6x^2+9x+1$ on $[0,4]$. **Answer: max** 5**, min** 1.
21. Inflection points of x^4-4x^3. **Answer:** $x=0$ **and** $x=2$ ($f''=12x(x-2)$ changes sign at both).
22. Minimum of $x+\frac{1}{x}$ for $x>0$. **Answer:** 2 **at** $x=1$.
23. Nature of $x=0$ for $f(x)=x^4$ and for $f(x)=x^3$. **Answer: minimum for** x^4 **(although** $f''(0)=0$**); neither max nor min for** x^3 **(inflection).**
24. Gradient descent on $f(x)=(x-3)^2$ from $x_0=0$ with $\eta=0.1$: x_1,x_2. **Answer:** 0.6 **and** 1.08.
25. One Newton step for $x^2-2=0$ from $x_0=1$. **Answer:** 1.5.

How to study this section: Spend most of your time on limits (standard limits, L'Hôpital, Taylor shortcuts), on the continuity versus differentiability relationships, on finding and classifying critical points, and on Taylor expansions of $e^x,\sin x,\cos x,\ln(1+x)$. Highest-yield for GATE DA: evaluating limits, parameter-matching for continuity, checking differentiability at corners, Taylor coefficients/approximations, and maxima/minima on closed intervals.